Uniqueness of a smooth real potential on a known Riemannian manifold

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Abstract

We prove uniqueness of a smooth real scalar potential from its full Dirichlet-to-Neumann map at one energy on an arbitrary known compact, connected, smooth Riemannian manifold of dimension at least three with smooth nonempty boundary. The only spectral assumption is that this energy is not a Dirichlet eigenvalue; negative eigenvalues and sign-changing potentials are allowed. Equal boundary maps first give common exterior source solutions through a weak gluing argument that does not require a positive quadratic form. Near an interior frontier, a common source gives local positive conjugations of the two equations. We adapt the separately elliptic continuation and shrinking-sphere method of OpenAI to the resulting operators with the same principal metric. An angular coercivity estimate bounds the actual transfer densities jointly in center and direction. Their collapse gives equality on one neighborhood for every matching source pair. Finite source jets then recover the local conjugating solutions and the potentials, closing the frontier without a global positive solution.

1 Introduction

Let (M, g) be a known compact connected smooth Riemannian manifold of dimension $n \geq 3$ with smooth nonempty boundary. For a real smooth potential q such that zero is not a Dirichlet eigenvalue of $L_q = -\Delta_g + q$, define the ordinary Dirichlet-to-Neumann map by

$$\Lambda_{g,q} f = \partial_{\nu_g} u|_{\partial M}, \qquad L_q u = 0 \text{ in } M, \qquad u|_{\partial M} = f, \tag{1}$$

where ν_g is the outward unit normal. The map acts from $H^{1/2}(\partial M)$ to $H^{-1/2}(\partial M)$; equivalently, its values and pairings on all smooth boundary data determine the map. The boundary is labeled, and both potentials are compared on the same manifold with the same metric. We ask whether this single boundary response determines the interior scalar coefficient without a geometric restriction on g.

The fixed-energy question must be distinguished from recovery using a family of energies or a sufficiently large frequency together with a ray transform hypothesis. For example, Ma, Sahoo and Salo [5] establish a large-fixed-frequency result on manifolds with an invertible ray transform. Such results do not give the assertion below on an arbitrary known smooth manifold at the prescribed energy zero. We do not assume simplicity, absence of conjugate points, a limiting Carleman weight, or injectivity of a geodesic ray transform.

Our continuation construction adapts the mixed-kernel transfer, product continuation, shrinking-sphere construction, and uniform density and coercivity estimates of OpenAI [6]. A separately elliptic mixed Green kernel is continued along product hypersurfaces, and its transfer integrals on

shrinking spheres are controlled through angular raising and lowering operators. The later metric and unitary-connection preprint [7] addresses an operator of the form $d_A^* d_A$; its theorem does not include an additional independent real scalar potential. We give the continuation and angular arguments needed here, with their overlap and parameter dependence specified, and adapt them to two locally conjugated scalar equations having the same principal metric. The core continuation method is credited to these sources.

Theorem 1.1. *Let (M, g) be any compact connected smooth Riemannian manifold of dimension $n \geq 3$ with smooth nonempty boundary. Let $q_1, q_2 \in C^\infty(M; \mathbb{R})$, and suppose zero is not a Dirichlet eigenvalue of either $L_j = -\Delta_g + q_j$, with Dirichlet domain $H^2(M) \cap H_0^1(M)$. If*

$$\Lambda_{g,q_1} = \Lambda_{g,q_2}$$

as full ordinary boundary maps at energy zero, then $q_1 = q_2$ on M.

No orientability or connectedness of the boundary is required. The metric is known throughout, and no diffeomorphism or coefficient gauge occurs in the conclusion. Both potentials are arbitrary real smooth functions subject to the stated nonresonance condition. In particular, either Dirichlet operator may have negative eigenvalues, and the difference of the potentials may be flat or change sign arbitrarily often.

The spectral qualification matters for the proof. A global positive solution s of $(-\Delta_g + q)s = 0$ would give, for interior compactly supported u,

$$\int_M (|\mathrm{d}u|_g^2 + q|u|^2)\mathrm{d}V_g = \int_M s^2 |\mathrm{d}(u/s)|_g^2 \mathrm{d}V_g \geq 0.$$

By density this excludes negative Dirichlet eigenvalues. Thus a global positive conjugation would impose a stronger assumption than the theorem. Shifting the operator to obtain positivity would also change the measured energy. We instead use nonresonance for the signed weak inverse and select positive source solutions only in a small neighborhood of a frontier.

Here is the structure of the argument. Equality of the boundary maps gives equal boundary jets of the potentials. A thin exterior cap can therefore be attached with common coefficients. Nonresonance persists for a small cap, and weak stationarity transfers every source supported in the cap between the two Dirichlet inverses. Point distributions and scalar unique continuation show that these sources have arbitrary local value and first jets subject only to their differential equation.

At a point where all source families agree from one side, one matching source is positive nearby. Dividing each equation by its own positive solution removes the zeroth-order term locally. A compatible mixed kernel then defines a transfer density on small spheres. Strict product hypersurfaces continue this kernel at every positive radius; no uniform kernel bound at radius zero is asserted. The complete radial map, including its smooth remainder, gives an equation for the actual density. Concavity of a reference depth function supplies an angular coercivity estimate and a uniform joint L^2 bound in the center and sphere variables.

The radii collapse on one fixed neighborhood. The joint bound makes every matching solution pair agree there, with that neighborhood chosen before the further source. A finite spanning set of source gradients then identifies the two local positive conjugations and the original potentials. This common-neighborhood conclusion advances the open set of agreement past any putative interior frontier and proves the theorem globally.

Section 2 establishes the signed inverse, cap data and source jets. Sections 3–5 construct the local conjugation and the continued mixed kernel. Sections 6 and 7 derive and bound the transfer density. Section 8 closes the global frontier. We use $\Delta_g = \mathrm{div}_g \nabla_g$, positive volume densities, and

the outward normal convention in (1). All coefficients are smooth up to any boundary on which they are used. The proof does not assert a partial boundary-data theorem, a complex-potential theorem, or a resonant case.

2 Signed Dirichlet inverses and common exterior data

We first obtain interior source data from the boundary response. This step must allow an indefinite Dirichlet form. Throughout, $\Delta_g = \operatorname{div}_g \nabla_g$, $L_j = -\Delta_g + q_j$, and all integrals on manifolds use positive densities. For real functions put

$$\mathfrak{b}_j(u,v) = \int_M (\langle \mathrm{d}u, \mathrm{d}v\rangle_g + q_j uv)\mathrm{d}V_g.$$

The complex extension is sesquilinear, linear in its first variable. The Dirichlet realization has domain $H^2(M)\cap H_0^1(M)$, where H_0^1 is the closure of interior compactly supported smooth functions. Smooth data suffice for every boundary pairing below.

Lemma 2.1. *If 0 is not a Dirichlet eigenvalue of $L = -\Delta_g + q$ with real smooth q, its weak realization $H_0^1 \to H^{-1}$ is a bounded isomorphism. This property persists under a sufficiently small smooth outward perturbation of the boundary and coefficients, pulled back to M.*

Proof. Add a constant $c > 0$ so that $\mathfrak{b}_c$ is coercive on H_0^1. Lax–Milgram gives its inverse R_c [3, Theorem 4.20]. The inclusion $H_0^1 \hookrightarrow L^2$ is compact: extend its defining approximants by zero in boundary charts, localize in finitely many cubes, and truncate Fourier frequencies. The H^1 bound makes the remaining L^2 norm $O(N^{-1})$. Thus $K = cR_cJ$, with J the L^2 pairing, is compact and self-adjoint in the $\mathfrak{b}_c$ inner product. A weak vector in $\ker(I-K)$ gains H^2 by boundary elliptic regularity and is a zero Dirichlet eigenfunction; hence this kernel is zero. If $I-K$ were not bounded below, a unit sequence and compactness of K would produce a nonzero kernel vector. Its range is therefore closed; its orthogonal complement is $\ker(I-K)^* = 0$, so it is onto. This proves the asserted inverse. Boundary regularity here is the pure divergence estimate of [3, Theorems 4.30–4.31]; smooth lower terms enter the forcing and are bootstrapped in finite charts.

After pullback, an outward perturbation changes the weak form in operator norm by a quantity tending to zero. If R is the original inverse, $\|R(\mathfrak{b}_\epsilon - \mathfrak{b}_0)\| < 1$ gives the perturbed inverse by a Neumann series. Positivity of the original form is unnecessary. □

Equality of the DN maps determines the complete boundary jets of q_1, q_2. For the identical metric this is the scalar, zero-connection case of the last clause of [2, Theorem 3.4]; the normal gauges are identity. To fix the sign, in inward boundary distance s let $\rho = (g^{ab}\xi_a\xi_b)^{1/2}$. The factor symbol starts with $b_1 = -\rho$, and b_0 depends only on g. Its scalar recursion gives

$$b_{-k-1,1} - b_{-k-1,2} = -\frac{\partial_s^k(q_1-q_2)}{(2\rho)^{k+1}} + \text{terms fixed by lower potential jets and metric jets}, \qquad k \ge 0. \tag{2}$$

Equal complete DN symbols recover these jets inductively. The simultaneous metric normalization in the first clause of that theorem is not needed when the metric is already identical. Flatness at the boundary gives jets, not equality on an interior collar.

Choose a boundary patch and a nonnegative smooth bump with connected positivity set. Attach a thin cap E across that patch, obtaining a smooth compact manifold N. Extend the metric smoothly and use one common exterior extension for the two potentials. This is possible by (2): their difference has a smooth zero extension. A smooth extension of either coefficient can be

constructed from cutoff Taylor monomials, choosing the jth support small enough that its derivatives of order less than j are bounded by 2^{-j}. The series converges in every fixed smooth norm. A collar diffeomorphism identifies N with a small boundary perturbation of M. Lemma 2.1 makes both extended Dirichlet problems invertible after the cap is made sufficiently thin.

Write $R_j = L_{j,D}^{-1}$ on N, and normalize its kernel by

$$R_j F(x) = \int_N G_j(x,y) F(y) \mathrm{d}V_g(y).$$

The kernels are real and symmetric. An interior order -2 parametrix Q with $L_j Q = I + S$ gives $G_j = Q - R_j S$ locally; S and $R_j S$ have smooth kernels. The off-diagonal symbol-kernel estimate [9, Lecture I, (1.21)–(1.24)] gives

$$|\partial_x^a \partial_y^b G_j(x,y)| \le C_{a,b} |x-y|^{2-n-|a|-|b|} \tag{3}$$

on sufficiently small compact coordinate neighborhoods, with a smooth remainder absorbed into the constant. Its order condition is satisfied because $n \ge 3$.

Lemma 2.2. *For every $F \in C_c^\infty(E)$, $R_1 F = R_2 F$ on E. Consequently $G_1 = G_2$ on $E \times E$.*

Proof. Let $w_1 = R_1 F$, and replace its restriction to M by the L_2 solution with the same boundary trace h. Leave it equal to w_1 on E. The replacement differs on M by a member of $H_0^1(M)$; its zero extension gives a glued member of $H_0^1(N)$. In the weak equation, the difference of the interior boundary pairings is $\langle (\Lambda_2 - \Lambda_1) h, \phi|_{\partial M} \rangle = 0$. The cap coefficients and forcing are common, so the glued function solves $L_2 w = F$. Uniqueness identifies it with $R_2 F$. This is a stationarity argument for the weak form, not minimization. Testing all cap sources and using smoothness off the pole gives the kernel identity; the equal delta singularities also cancel distributionally. □

Fix a nonempty source set $U_0 \Subset E$.

Lemma 2.3. *At any $p \in N^\circ \setminus \overline{U_0}$, the second jets of $R_j F$, $F \in C_c^\infty(U_0; \mathbb{R})$, fill exactly the hyperplane $L_j v(p) = 0$. In particular, value and first derivatives are arbitrary.*

Proof. Let an order-at-most-two point distribution T annihilate these jets. Define $R_j T$ by transposition. Then $R_j T = 0$ on U_0 and $L_j R_j T = T$. Away from p it is smooth by a local parametrix and vanishes by scalar unique continuation [1]. The punctured interior is connected in dimension at least two: paths can detour around a point in a coordinate ball. Thus $R_j T$ is supported at p.

A point-supported distribution is a finite sum of delta derivatives. Indeed, a test whose derivatives through its finite order vanish at p can be cut off at radius ϵ; Taylor's remainder makes its test seminorm $O(\epsilon)$ and its pairing vanish. For a nonzero such distribution of order k, an elliptic second-order operator has exact image order $k+2$: its top polynomial is multiplied by a nonzero quadratic principal symbol. Hence $R_j T = c\delta_p$ and $T = c L_j \delta_p$, where distribution transposes use $\mathrm{d}V_g$. This is precisely the annihilator of the equation hyperplane. Finite-dimensional duality proves surjectivity onto it. The Hessian trace can enforce the equation while value and first derivatives are prescribed. Real parts give real sources. □

3 Local positive conjugation

At a point where the two source families agree from one side, choose a matching pair with common value one using Lemma 2.3. On a fixed small neighborhood it gives positive solutions s_j of $L_j s_j = 0$. For $u_j = s_j v_j$ the exact conjugation is

$$s_j^{-1} L_j (s_j v_j) = -\Delta_g v_j - 2\langle \mathrm{d}\log s_j, \mathrm{d}v_j \rangle_g =: A_j v_j = -s_j^{-2} \operatorname{div}_g (s_j^2 \nabla_g v_j). \tag{4}$$

Thus $A_j 1 = 0$ and both principal metrics are identical. Their local symmetric densities are $s_j^2 \mathrm{d}V_g$, and

$$K_j(x,y) = \frac{G_j(x,y)}{s_j(x)s_j(y)}$$

is the corresponding local Green kernel. Only local positivity is needed.

The following statement is the local result used in the global proof. The kernels in it are the actual kernels just constructed on N.

Proposition 3.1. *Suppose a neighborhood Ω of 0 has a known open subset W containing one side of a smooth hypersurface through 0. Suppose $s_1, s_2 > 0$ solve $L_j s_j = 0$ on Ω, agree on W, and $G_1 = G_2$ on $W \times W$. There is one neighborhood Z of 0 such that every pair of smooth solutions $A_j v_j = 0$ on Ω agreeing on W satisfies $v_1 = v_2$ on Z.*

We prove this through a transfer integral on shrinking spheres. The neighborhood Z and the integral density will depend on the fixed kernels, groundings and geometry, but not on a further solution pair. Choose coordinates with $g(0) = I$ and known side tangent to $x_n < 0$. Under $x \mapsto \lambda x$, set

$$g_\lambda(x) = g(\lambda x), \quad s_{j,\lambda}(x) = s_j(\lambda x), \quad K_{j,\lambda}(x,y) = \lambda^{n-2} K_j(\lambda x, \lambda y).$$

On fixed bounded sets the principal metric tends smoothly to Euclidean, and the grounded drift tends to zero. Each compact subset strictly below the plane becomes known for small λ. Estimate (3) gives uniform smooth bounds on compact separated pairs. Initially use K_1 when y is known and K_2 when x is known. They agree on the intersection and satisfy $A_{1,x} K = A_{2,y} K = 0$ off the diagonal. For the rest of the local proof subscripts λ are suppressed. The small spatial scale will be fixed before any radius tends to zero.

4 Separately elliptic continuation

We give the continuation argument since its domain and overlap properties are used below. Scalar local regularity, quantitative interior estimates, and the calculus used here follow from an elliptic parametrix [9, Lecture I, Sections 3–4]. On compact buffers the required integer estimates have finite coefficient dependence. For example, testing $Pu = \partial_i F^i + h$ against $\eta^2 \bar{u}$ gives

$$\|\nabla u\|_{L^2(U)} \le C(\|u\|_{L^2(V)} + \|F\|_{L^2(V)} + \|h\|_{L^2(V)}), \qquad U \Subset V.$$

Differentiating a homogeneous equation gives divergence forcing involving only already controlled derivatives, and finite nested buffers give every integer order. Initially distributional L^2 solutions gain local Sobolev regularity by the parametrix, so the weak tests are legitimate. Constants use only finitely many smooth coefficient bounds at each specified order.

4.1 Strict scalar weights

Let θ have nonzero differential and, with $Z = \nabla\theta/|\mathrm{d}\theta|$, suppose

$$\operatorname{Hess}\theta(Z,Z) + \operatorname{Hess}\theta(V,V) > 0 \quad (V \perp Z,\ |V| = 1). \tag{5}$$

On a small neighborhood choose m strictly between minus the least tangential Hessian eigenvalue and the normal Hessian value. For $v = e^{\tau\theta} w$, the conjugated positive Laplacian is

$$A_\tau + B_\tau = -\Delta - \tau^2 |\mathrm{d}\theta|^2 + \tau(2\nabla\theta \cdot \nabla + \Delta\theta),$$

where A_τ is self-adjoint and B_τ skew-adjoint on compact supports. Integration by parts gives

$$\|(A_\tau + B_\tau)v\|^2 = \|A_\tau v\|^2 + \|B_\tau v\|^2 + \langle [A_\tau, B_\tau]v, v\rangle,$$
$$\langle [A_\tau, B_\tau]v, v\rangle = 4\tau \int \operatorname{Hess}\theta(\nabla v, \nabla \bar{v}) + 4\tau^3 \int \operatorname{Hess}\theta(\nabla\theta, \nabla\theta)|v|^2 - \tau \int \Delta^2\theta |v|^2.$$

Ricci terms from derivative commutation cancel those in $\operatorname{div}\operatorname{Hess}\theta = \mathrm{d}\Delta\theta + \operatorname{Ric}(\nabla\theta, \cdot)$. Add $4m\tau \operatorname{Re}(A_\tau v, v)$ to this form. It makes the tangential gradient and leading scalar terms positive. The remaining normal-gradient loss is controlled by $\|\nabla_Z v\|^2 \le C\tau^{-2}\|B_\tau v\|^2 + C\|v\|^2$. The added form costs at most $\frac{1}{4}\|A_\tau v\|^2 + C\tau^2\|v\|^2$. For large τ these losses are absorbed. Smooth lower terms have conjugated norm squared at most $C(\|\nabla v\|^2 + \tau^2\|v\|^2)$ and are absorbed too. For the actual scalar operator A this yields

$$\tau\|e^{\tau\theta}\mathrm{d}w\|^2 + \tau^3\|e^{\tau\theta}w\|^2 \le C\|e^{\tau\theta}Aw\|^2, \qquad \tau \ge \tau_0. \tag{6}$$

The constants are uniform on fixed strict margins and finite coefficient bounds.

Suppose a cutoff equals one on $O \Subset D$, its bottom derivative support has weight at most a, the remaining derivative support at most b, and $\theta \ge c$ on O, where $a < c < b$. Enlarge the latter support to a compact observed open set $S \Subset D$. Interior estimates on the enlarged collars and (6) give, for $Au = 0$,

$$\|u\|_O \le C\{e^{-\tau(c-a)}\|u\|_D + e^{\tau(b-c)}\|u\|_S\}.$$

Balancing the terms when the balancing τ exceeds τ_0, and using the trivial bound otherwise, proves

$$\|u\|_O \le C\|u\|_S^{\gamma}\|u\|_D^{1-\gamma}, \qquad \gamma = \frac{c-a}{b-a}. \tag{7}$$

If the observed norm is zero, let $\tau \to \infty$.

4.2 Extension from a cross

Lemma 4.1. *Let D_i be bounded connected coordinate domains and $S_i, O_i \Subset D_i$ open, with S_i nonempty. Let A_i be smooth real scalar elliptic operators with positive principal matrices. Suppose their L^2 solutions satisfy (7) with exponents $0 < \gamma_i < 1$ and $\alpha = \gamma_1 + \gamma_2 - 1 > 0$. Compatible separately homogeneous L^2 functions on $(D_1 \times S_2) \cup (S_1 \times D_2)$ extend to a smooth separately homogeneous function on $O_1 \times O_2$, agreeing on its actual intersections with both given pieces. On compact output subsets its smooth norms are bounded by the summed input L^2 norms.*

Proof. Let $\mathcal{H} = \{w \in L^2(D_1) : A_1 w = 0\}$, a closed Hilbert subspace. Restriction $R : \mathcal{H} \to L^2(S_1)$ is compact by cutoff Fourier truncation and injective by unique continuation. On each fixed cutoff coordinate cube, identify opposite faces and use the torus Fourier basis [8, Section 2, Theorem 21 and Corollary 22]. Parseval and the integer derivative multipliers give, for each integer m, a rank-$O(N^{n_1})$ approximation with norm error $O(N^{-m})$, where $n_1 = \dim D_1$.

Here the compact spectral construction can be made directly. Maximize $\|Rw\|^2$ on the unit sphere using a weakly convergent maximizing sequence and compactness of R. The maximizing vector has norm one; variation in orthogonal directions makes its restriction orthogonal to all subsequent ones. Iterate on orthogonal complements. The resulting basis w_j is complete: a vector perpendicular to all of it has image norm bounded by every decreasing singular value and hence zero. In the infinite case compactness forces these values to zero. Thus

$$\|w_j\|_{D_1} = 1, \quad (Rw_i, Rw_j)_{S_1} = \mu_j^2\delta_{ij}, \quad 0 < \mu_j \le 1.$$

A vector in the span of the first j basis vectors annihilated by a rank-less-than-j approximation has image norm at least μ_j. The Fourier error therefore gives $\mu_j \le C_m j^{-m/n_1}$, and in particular decay faster than any fixed power. No lower bound is used. The same argument covers finite-dimensional spaces.

Write the first cross piece $F_1 = \sum_j w_j b_j^0$ on $D_1 \times S_2$. For the second piece define

$$b_j(y) = \mu_j^{-2} \int_{S_1} F_2(x,y)\overline{w_j(x)}\mathrm{d}x.$$

If the summed input norm is M, Cauchy–Schwarz gives $\|b_j\|_{D_2} \le M/\mu_j$ and compatibility gives $b_j = b_j^0$ on S_2, $\|b_j\|_{S_2} \le M$. Fubini and compact-test approximation in x give $A_2 b_j = 0$. Interpolation then gives

$$\|w_j\|_{O_1} \le C\mu_j^{\gamma_1}, \quad \|b_j\|_{O_2} \le CM\mu_j^{\gamma_2 - 1}, \quad \|w_j b_j\|_{O_1 \times O_2} \le CM\mu_j^{\alpha}.$$

The series converges absolutely in product L^2 and solves both equations. Their sum is elliptic on the product, giving smoothness and compact-subset estimates.

To identify the second piece, put $V = \overline{R\mathcal{H}}$. For $z \perp V$, the A_2 solution $\int_{S_1} F_2(x,y)\bar{z}(x)\mathrm{d}x$ vanishes on S_2 and hence on D_2. A countable dense set of such z proves that almost every second-piece slice belongs to V. Parseval in the basis Rw_j/μ_j identifies its coefficients as $\mu_j b_j$. The same partial sums converge to F_2 on $S_1 \times D_2$, and absolutely on $S_1 \times O_2$ by the bound $CM\mu_j^{\gamma_2}$. They agree with the product limit on $(S_1 \cap O_1) \times O_2$. Completeness in $\mathcal{H}$ similarly gives agreement with F_1 on $O_1 \times (S_2 \cap O_2)$. For a complete orthonormal basis, Pythagoras and the limit of orthogonal projections give Parseval. Its squared tails pass through the finite product integrals by dominated convergence, bounded by the original squared L^2 norm. The interchanges use dominated convergence and absolutely integrable Fubini [4, Theorems 18 and 23]. If $\mathcal{H} = 0$, the first piece is zero and the same pairings plus unique continuation make the second piece zero.

For uniform families choose a fixed positive lower bound for α, fixed buffers, uniform interpolation constants and finite coefficient bounds. One Fourier decay order then makes $\sum_j \mu_j^{\alpha}$ uniformly bounded. Smooth output estimates are uniform under these same bounds. □

4.3 Crossing a product hypersurface

Let $\rho = 0$ be a smooth product hypersurface with both partial gradients nonzero. Put $a_i = \nabla_i\rho/|\mathrm{d}_i\rho|^2$ and $H = \operatorname{Hess}\rho$ in the product principal metric. A sufficient strict condition is

$$H((a_1,-a_2),(a_1,-a_2)) + H((v_1,v_2),(v_1,v_2)) > 0, \quad v_i \perp a_i, \quad |v_i| = |a_i|. \tag{8}$$

Lemma 4.2. *Under (8), a smooth separately homogeneous function on $\rho > 0$ extends across the point to a product neighborhood, agreeing with its original germ on the positive side.*

Proof. Let $\mathcal{C}_i$ consist of symmetric forms P_i with $P_i(a_i,a_i) + P_i(v_i,v_i) > 0$ for every indicated v_i. Its dual consists of $d_i a_i a_i^T + S_i$, with $d_i \ge 0$, $S_i \ge 0$ tangential and $\operatorname{tr} S_i = d_i|a_i|^2$. The dual of $(\mathcal{C}_1 \oplus \mathcal{C}_2) + \mathrm{PD}$ consists of positive semidefinite matrices with these diagonal blocks. Its trace-one slice is compact. If a dual matrix Q annihilates $\mathrm{d}\rho$, covariance decomposition makes the two normal components opposite, with common variance $d > 0$ and no covariance with tangential components. The tangential covariance has two fixed traces. Every such covariance is a finite convex combination of rank-one ones: at rank $r \ge 2$, a supported symmetric variation preserving the two traces exists since $r(r+1)/2 > 2$. Moving in both directions to the first singular endpoints lowers rank and expresses the original as their convex combination; repeat. The rank-one covariances are $d(v_1,v_2)(v_1,v_2)^T$. Thus (8) is positivity of $\operatorname{tr}(QH)$ on all annihilating dual tests.

Compactness gives a positive margin on those tests. Near them it persists; away from them $\operatorname{tr}(Q\mathrm{d}\rho\otimes\mathrm{d}\rho)$ has a positive lower bound. Choose K large. Finite-dimensional convex separation then gives

$$H + K\mathrm{d}\rho\otimes\mathrm{d}\rho > P_1 \oplus P_2, \qquad P_i \in \mathcal{C}_i.$$

The separation used here follows from nearest-point projection onto a closed convex set, followed by a limit of supporting normals at a boundary point. Choose factor functions θ_i with first jets $\mathrm{d}_i\rho$ and Hessians P_i, and value zero at contact. Taylor's formula gives

$$\theta_1 + \theta_2 \le \rho + K\rho^2/2 - c(|x|^2 + |y|^2) \tag{9}$$

on a small product neighborhood.

Take factor cylinders of lateral radius R and height $(-l, l+4e)$, with $l \ll cR^2$ and $e \ll \kappa l^2$. A cutoff is one on the inner lateral cylinder for $-l+3e < \theta_i < l+2e$; its top derivatives lie in $l+2e < \theta_i < l+3e$. Observe the enlarged band $l+e < \theta_i < l+7e/2$ and an enlarged lateral collar. The top observed cross has height sum greater than e; the lateral cross has quadratic gap larger than the possible negative height sum. By (9) both crosses lie strictly on $\rho > 0$.

Use the strict scalar weights $\phi(\theta_i) = \theta_i - \kappa\theta_i^2$. Choose $\kappa > 0$ small to preserve (5). The bottom, top derivative and target weights can be taken as

$$a = \phi(-l+3e), \quad b = \phi(l+3e), \quad c = \phi(-e).$$

The output has inner lateral radius and $-e < \theta_i < l+2e$. Equation (7) gives $\gamma_i = (c-a)/(b-a) > 1/2$ for small $e > 0$, since at $e = 0$ it equals $1/2 + \kappa l/2$. Lemma 4.1 therefore applies. The enlarged observed band intersects the output in the open band $l+e < \theta_i < l+2e$; its norm enlargement does not change the derivative weight b. Near contact, increasing the second height to $l+3e/2$ with other coordinates fixed stays in the output and in $\rho > 0$: $\partial_{\theta_2}\rho = 1$ at contact and remains positive nearby. At the endpoint the new and old functions agree on their actual cross intersection. Second-variable unique continuation along this path identifies the near-contact positive-side germs. The factors can be exchanged. All choices have fixed positive margins and persist under small smooth coefficient perturbations. □

4.4 A compatible kernel at fixed separation

For a plane point x_0 and separation scale s, choose $\tau = 1/100$. Use first domain $B(x_0 - \tau s e_n, 3\tau s)$ and observed radius $(1-\alpha)\tau s$. In the second variable use $s/2 < |y - x_0| < 2s$ and a small observed ball about $x_0 - se_n$. These domains are separated. Radial weights $|z - z_*|^{-1}$ have normal Hessian $2|z-z_*|^{-3}$ and tangential value $-|z-z_*|^{-3}$, so they are strict. A finite annular ball chain gives a fixed second exponent $\gamma_2 > 0$. In the first ball, as α and a positive output excess cs decrease, $\gamma_1 \to 1$. Choose them so $\gamma_1 + \gamma_2 > 1$. The cross lemma reaches the product of a ball about x_0 with a narrower annulus about radius s.

Retain plane balls $|x - x_0| < cs/1000$ and smaller annuli, leaving comparison margins inside the larger output products. On overlaps the scales are comparable. Lowering the near-plane variable by a small multiple of c times the smaller scale stays in both larger first balls and in their annular margins. Choose η_0 smaller than these margins and than $\alpha\tau s/4$ over the compact scale range. For each output, first compare with its own original prescription: first-variable unique continuation from its observed ball identifies it with K_2 on the connected lower part $x_n < -\eta_0$ of its output. Second-variable continuation similarly identifies it with K_1 on the lower part of its annulus. Those compact sublevels are known for small λ. In an overlap of the same orientation, lower its near-plane variable to this region. In an opposite-orientation overlap, lower both variables

by $\max(x_n, y_n, 0) + 2\eta_0$. Their difference is unchanged, and the retained widths keep the paths inside both larger outputs. Both germs then equal the actual common kernel. Unique continuation identifies them on each overlap component. We have one kernel on the coarse region

$$\min(x_n, y_n) < c_0|x - y| \tag{10}$$

for a fixed $c_0 > 0$ on every fixed compact range of locations and positive separations.

To extend further, use tubes $x = y + r_a(y)\omega$, $|\omega| = 1$, where

$$t = M + y_n + e_0|y'|^2, \qquad r_a = at^{1+\sigma}, \qquad y_n \geq -B - 1, \quad t \leq T_0.$$

Choose $M > B + 3$ and small $\sigma > 0$. Take $a_{\min} > 0$ small enough to pass the desired separations and $a_{\max}$ large enough to put its level in (10). Take $c_0 a_{\min} T_0^{\sigma} > 1$ and choose $e_0 > 0$ last. The base is compact; its bottom and $t = T_0$ faces are prescribed. Every tube has positive minimum radius.

In the Euclidean limit set $\rho = (|x - y|^2 - r_a^2)/2$, $b = \nabla r_a$, and

$$U_1 = \omega + iv, \ v \perp \omega, \ |v| = 1, \qquad U_2 = \frac{\omega + b}{|\omega + b|^2} + iw, \quad w \perp (\omega + b), \ |w| = |\omega + b|^{-1}.$$

Multiplying (8) by r_a^2 gives

$$|\pi_{\omega^\perp}(U_1 - U_2)|^2 - (r_a \operatorname{Hess} r_a)(U_2, \overline{U_2}). \tag{11}$$

Away from fixed neighborhoods of $b = -\omega, -2\omega$, the first term is at least $c|b|^2$ on bounded b ranges. Indeed its real part is $-b_\perp/|\omega + b|^2$; if this vanishes, b is parallel to ω, and the imaginary projected length equals $1/|\omega + b|$. The only zeros are $b = 0, -2\omega$. Near zero the real square controls $|b_\perp|^2 + O(|b|^3)$, and the imaginary length difference controls $(\omega \cdot b)^2 + O(|b|^3)$, uniformly over the permitted v, w. Compactness gives the remaining lower bound. Outside (10), $r_a/t \leq 1/c_0$, so b is bounded. The two excluded neighborhoods are already prescribed: with ∇t close to e_n, $y_n/r_a + \omega_n \leq (1 + \sigma)|\nabla t|/|b| + \omega_n$ is small near $-\omega$ and negative near -2ω. Finally

$$r_a|\operatorname{Hess} r_a| \leq C(\sigma + T_0 e_0)|b|^2.$$

Choose the exclusion sizes, then σ, then T_0, and lastly e_0 to absorb this error. Strictness persists for small spatial dilation.

Here is the compatibility argument used in this and the next continuation. The prescribed region is upward closed in a. Let a_* be a putative last reached level. Each local crossing there uses compact data strictly above $a_* + \eta_i$. A finite cover of the level gives outputs containing base boxes times intervals $(a_* - h_i, a_* + h_i)$. Every overlap has vertical segments to a common height above a_*, where both germs equal the preceding function. Unique continuation on neighborhoods of these segments identifies the germs on every component. The finite cover crosses a_*, a contradiction. This argument uses bounds on compact data above the last level, not a bound up to that level. It supplies a single compatible kernel on every desired compact separated region. To compare with the entire original cross, work in a larger region and remove a small interior closed ball about a known y from the first-variable ball. The remaining domain is connected for $n \geq 2$ and contains a lower open ball where the kernels agree. First-variable unique continuation identifies them; exchange variables for the other prescription.

5 Shrinking spheres in a concave reference metric

On the rescaled coordinate cylinder $Y = \{|y'| \le 1, -1 \le y_n \le 1\}$ set $t_0 = y_n + |y'|^2$. Choose L large and put

$$\gamma = e^{-2Lt_0} g_\lambda, \qquad t(y) = \int_{1/8}^{t_0(y)} e^{-2Ls} \mathrm{d}s.$$

The conformal connection formula gives

$$\mathrm{Hess}_\gamma t = e^{-2Lt_0}\{\mathrm{Hess}_{g_\lambda} t_0 - L|\mathrm{d}t_0|^2_{g_\lambda} g_\lambda\} \le -c\gamma \tag{12}$$

after shortening the spatial range. In particular $\mathrm{d}t \neq 0$ and $t(0) < 0$. Choose p large, then $R_* > 0$ small, and define for $0 < \delta \le 1$

$$\ell_\delta(t) = \delta \log(1 + e^{t/\delta}), \quad r_\delta = R_* \ell_\delta^p, \quad f_\delta = \log r_\delta, \quad \beta_\delta = |\mathrm{d}f_\delta|_\gamma. \tag{13}$$

The function $a(s) = e^s/\{(1+e^s)\log(1+e^s)\}$ is positive and decreasing; it tends to one at $-\infty$ and is comparable to $(1+s)^{-1}$ at $+\infty$, with $|a^{(j)}|/a^{j+1}$ bounded for each fixed j. Thus (12) gives, uniformly in δ,

$$\beta \ge \beta_0, \qquad \mathrm{Hess}_\gamma f \le -c\beta\gamma, \qquad |\nabla^j f| \le C_j \beta^j, \qquad r\beta^2 \le \varepsilon. \tag{14}$$

Here β_0 is made large by p, and the last bound follows from $r\beta^2 = R_* p^2 \ell^{p-2} (\ell')^2 |\mathrm{d}t|^2$ for $p \ge 2$. The radii strictly increase with δ, since $\partial_\delta \ell_\delta = \log(1+e^s) - s e^s/(1+e^s) > 0$. They tend uniformly to zero on compact subsets of $t < 0$.

Multiplying the grounded equations by e^{2Lt_0} gives

$$\widetilde{A}_j = -\Delta_\gamma - 2\langle \mathrm{d}\log\sigma_j, \mathrm{d}(\cdot)\rangle_\gamma, \qquad \sigma_j = e^{(n-2)Lt_0/2} s_j. \tag{15}$$

The symmetric density is $\sigma_j^2 \mathrm{d}V_\gamma = e^{-2Lt_0} s_j^2 \mathrm{d}V_g$. Its reciprocal scalar change normalizes the delta exactly, so the kernels K_j and physical weighted fluxes are unchanged. We use these normalized operators below and again denote them by A_j.

For a sphere $d_\gamma(x,y) = r(y)$ use $\rho = (d_\gamma(x,y)^2 - r(y)^2)/2$ and identify endpoints by parallel transport. Put $b = \nabla r = r\nabla f$. The criterion vectors are the U_1, U_2 above, and their denominators are bounded because $|b| \le \varepsilon/\beta$. The short-geodesic Jacobi equation has curvature coefficient $O(r^2)$ on parameter interval $[0,1]$. Integrating twice with fixed endpoint values and using its index form gives distance Hessian $|U_1 - U_2|^2 + O(r^2)$, with fixed geometric constants. Consequently the strict expression is

$$|\pi_{\omega^\perp}(U_1 - U_2)|^2 - |b \cdot U_2|^2 - r^2 \,\mathrm{Hess}\, f(U_2, \overline{U_2}) + O(r^2).$$

The first term is at least $|b|^2 - C|b|^3$. The orthogonal real and imaginary vectors in U_2 have equal length $|\omega + b|^{-1}$, so $|b \cdot U_2|^2 \le |b|^2/|\omega + b|^2$. Concavity leaves

$$cr^2\beta - Cr^2 - Cr^3\beta^3 > 0 \tag{16}$$

for large p and small ε. There is no principal-metric mismatch term.

Fix any $\delta > 0$. The spheres with $\delta \le \delta' \le 1$ form a compact off-diagonal cylinder. The initial level has positive minimum radius. Choose $t_b > 0$ with $3t_b < t(1/4)$. On $t > t_b$ all radii are at least $R_* t_b^p$, so the fixed-separation kernel prescribes them. On $y_n < -1/4$ the actual K_1 prescribes them. These regions cover the base faces and are upward closed in δ'. The vertical-box compatibility argument in Section 4.4, using Lemma 4.2 and (16), extends one kernel near the entire family. Distinct positive lower endpoints agree by the same continuation from the common upper region.

Thus there is one compatible mixed kernel near every positive-radius sphere. This step asserts no kernel bound uniform at radius zero.

Take a fixed cutoff χ compactly supported in Y°, equal to one near 0, whose derivatives are supported in

$$\{y_n < -1/4\} \cup \{t > t_b\}. \tag{17}$$

For example multiply a cutoff rising between $y_n = -3/4, -1/2$ by one falling between $t = 2t_b, 3t_b$. The depth bounds keep the lateral support inside $|y'| < 1$.

6 The actual transfer density and its equation

For a first-side solution near the closed ball, define

$$\mathcal{T}_y v = \int_{\partial B_\gamma(y,r(y))} \sigma_1^2 \{K(x,y)\partial_{\nu_\gamma} v(x) - v(x)\partial_{\nu_\gamma} K(x,y)\} \mathrm{d}S_\gamma(x). \tag{18}$$

The first Dirichlet radial operator on this small ball moves to the smooth kernel trace by sphere adjunction. Pulling back to the unit tangent sphere S_y with probability measure $\mathrm{d}\omega_y$ gives a smooth density

$$\mathcal{T}_y v = \int_{S_y} \psi(y,\omega) v(\exp_y^\gamma(r(y)\omega)) \mathrm{d}\omega_y. \tag{19}$$

All surface Jacobians and weights are included in ψ. For y near a fixed center replace the moving boundary by a fixed slightly larger surrounding surface lying in the kernel's open annulus. Green's identity preserves the flux. The second operator can then be applied under that fixed integral: $A_{2,y}\mathcal{T}_y v = 0$. On the lower known set the actual Green formula makes $\mathcal{T}_y v = v(y)$. Unique continuation on connected Y gives

$$\mathcal{T}_y 1 = 1, \qquad \mathcal{T}_y v_1 = v_2(y) \tag{20}$$

for every matching local pair. In particular $\psi - 1$ has zero fiber mean.

We need the complete smooth parameter dependence of the small-ball radial map, including its smoothing part. Here and below Ψ^m means scalar classical pseudodifferential operators of order m on the unit sphere, with local quantization $(2\pi)^{-(n-1)} \int e^{i(\omega - z)\xi} a(\omega,\xi) u(z) \mathrm{d}z \mathrm{d}\xi$.

Lemma 6.1. *In smooth orthonormal center frames, the fixed-unit-ball Schrödinger radial map is $D_{\mathrm{Sch}}(y,r) = B + r^2 D_2(y,r)$, where B is the Euclidean radial DN map and D_2 is bounded and smooth in the complete Ψ^1 topology down to $r = 0$. The grounded radial map is*

$$D_{\mathrm{gr}} = B + r([B,m] - m) + r^2 \widetilde{D}_2, \qquad m = \mathrm{d}\log\sigma_1(y)\cdot\omega, \tag{21}$$

with the same smooth remainder property.

Proof. The unit-ball metric is $g_{y,r} = r^{-2}\exp_y(r\cdot)^*\gamma = I + O(r^2)$. Conjugation by σ_1 gives $-\Delta_{g_{y,r}} + r^2 Q(\exp_y(rz))$, with $Q = \Delta_\gamma \sigma_1/\sigma_1$. The Dirichlet form is uniformly coercive for small r by the unit-ball Poincaré inequality. Boundary integer regularity and the identity $R_a - R_b = R_a(L_b - L_a)R_b$, followed by $\partial R = -R(\partial L)R$, give smooth inverse dependence in every integer norm on the fixed domain.

The boundary factor recursion [2, Lemma 3.2 and Proposition 3.3] has denominator twice the positive tangential root, uniformly bounded below by $c|\xi|$. Its recursive symbols and all fixed parameter derivatives therefore have uniform symbol bounds. To include the smoothing part, write

inward distance $s = h\tau$, $h = |\xi|^{-1}$. The conjugated boundary operator times h^2 has expansion $-\partial_\tau^2 + \lambda^2 + \sum_{j\ge 1} h^j Q_j$, where $\lambda \ge c$ and Q_j have polynomial τ coefficients. Start with $a_0 = e^{-\lambda\tau}$; recursively solve

$$(-\partial_\tau^2 + \lambda^2)a_j = -\sum_{l=1}^{j} Q_l a_{j-l}, \quad a_j(0) = 0, \quad a_j \to 0.$$

Its forcing is a polynomial times $e^{-\lambda\tau}$. Writing $a_j = e^{-\lambda\tau}P_j$, solve $2\lambda P_j' - P_j''$ from highest polynomial degree downward and set $P_j(0) = 0$. Exponential decay bounds all fixed normal and parameter seminorms. High-frequency cutoffs tending sufficiently rapidly to infinity sum these symbols with arbitrarily small tails in each remainder topology. Taylor remainders in $s = h\tau$ and cutoffs away from the boundary are smoothing, uniformly with parameter derivatives. Finite chart patches give a Poisson parametrix H with exact trace after a smooth collar correction and with LH a smooth-kernel error. The true Poisson operator is $H - RLH$. Integer estimates on R give bounds from every negative boundary Sobolev order to every interior integer order for this correction and its parameter derivatives. These imply all smooth-kernel seminorms. For each fixed integer order $k \ge 1$, extend in flattened charts by finite reflections with coefficients c_ℓ satisfying $\sum_{\ell=1}^{k+1} c_\ell(-\ell)^m = 1$ for $0 \le m < k$. The finite Vandermonde system matches the required normal derivatives and gives a bounded H^k extension. Apply Fourier Cauchy–Schwarz to both kernel variables. Thus the true radial map is a complete smooth Ψ^1 family, not just a pointwise family modulo unspecified smoothing operators.

At $r = 0$ the map is B. Metric, potential and radial vector have zero first r variation, so its first variation is zero. Taylor's integral remainder proves the first assertion. On the boundary $w = \sigma_1(\exp_y(r\omega))/\sigma_1(y) = 1 + rm + O(r^2)$ and $\partial_\rho w = rm + O(r^2)$. Hence $D_{\mathrm{gr}} = w^{-1}D_{\mathrm{Sch}}w - w^{-1}\partial_\rho w$ gives (21). Its linear term has angular order zero. □

Let ∇_i denote horizontal Levi–Civita differentiation on the unit sphere bundle, preserving probability measure and acting covariantly on all displayed frame indices. We use orthonormal center frames for the fiber calculations; contracted formulas are independent of this choice. Let A_i differentiate the Euclidean harmonic extension in direction i, and take its fiber adjoint A_i^*. On degree-k harmonics $B = k$, $[B, A_i^*] = A_i^*$ and $\sum_i A_i^* A_i^* = 0$. The center-variation Jacobi field with transported direction has value e_i and initial derivative zero; its endpoint is parallel $e_i + O(r^2)$. Therefore the unit-ball velocity is $r^{-1}(e_i + O(r^2)) + f_i\omega$. Using the radial map for its normal component and taking the fiber adjoint of the boundary differentiation gives

$$C_i^* = r^{-1}A_i^* + f_i B + E_i + F_i, \qquad E_i \in r\Psi^1, \quad F_i \in \Psi^0. \tag{22}$$

The order-zero terms include $\omega_i([B, m] - m)$ and $r f_i([B, m] - m)$. The latter is bounded because $r\beta$ is bounded. All fixed base derivatives of the normalized coefficients of order j cost at most $C_j\beta^j$ by (14) and the complete parameter bounds of Lemma 6.1.

Write the second normalized equation locally as $-\gamma^{ij}\nabla_i\nabla_j - b_2^i\nabla_i$; b_2 includes the fixed coordinate/connection terms. Differentiating the transfer pairing gives

$$P\psi = 0, \qquad P = r\{\gamma^{ij}(\nabla_i + C_i^*)(\nabla_j + C_j^*) + b_2^i(\nabla_i + C_i^*)\}. \tag{23}$$

This tests every smooth boundary trace: solve the first Dirichlet problem on radius $r + \eta$ with a transported smooth trace h. Its restriction to the inner sphere tends to h in every fixed smooth norm as $\eta \downarrow 0$, by fixed-ball boundary regularity and smooth inverse dependence. These solutions are homogeneous near the closed inner ball. Thus the equation holds against every h, without imposing an ill-posed smooth Cauchy problem.

Put $D_i = \nabla_i + f_i B$ and

$$T = \sum_i A_i^* D_i, \qquad P = 2T + U + J. \tag{24}$$

The identities $D_i(r^{-1}A_j^*) = (r^{-1}A_j^*)D_i$ and $\sum_i (r^{-1}A_i^*)^2 = 0$ cancel the apparent r^{-2} terms. Here J is uniformly Ψ^1 with no base derivative, and U has base order $a = 0, 1, 2$ with coefficients

$$U_a \in \varepsilon \beta^{-a} \Psi^{2-a}. \tag{25}$$

A base derivative of order j has the extra allowance β^j. To see every term, put $R_i = r^{-1}A_i^*$ and $H_i = E_i + F_i$. The remaining products are rDD, rRE, rER, rDE, rED, rRF, rFR, rDF, rFD, rHH and the smooth drift terms. The rRF, rFR and rb_2R terms have bounded angular order one and enter J. The others have respectively second-base coefficient $r\gamma^{ij}$, first-base sizes $r\beta\Psi^1, r\Psi^0, r^2\Psi^1$, or zeroth-base sizes $r\beta^2\Psi^2, r\Psi^2, r^2\beta\Psi^2, r\beta\Psi^1$ and lower orders. All fit (25) using $r\beta^2 \le \varepsilon$. The second-base coefficient is real and independent of ω.

7 Angular coercivity and the joint density bound

For a fiber function use the harmonic decomposition $\phi = \sum_{k\ge 0} \phi_k$ and the spectral norm $\|\phi\|_s^2 = \sum_k (1+k)^{2s} \|\phi_k\|^2_{L^2(S_y)}$. Harmonic polynomials are complete: Fischer decomposition gives polynomial restrictions, and multivariate Bernstein approximation makes these dense in continuous sphere functions, then in L^2. The integer norms through order three agree with chart Sobolev norms by the sphere Laplacian parametrix; its eigenvalue on degree k is $k(k+n-2)$. For a scalar angular $Q \in \Psi^m$, $m = 1, 2, 3$, its two commutators with B still have order m. Integer mapping and adjoint bounds give

$$\|\Pi_k Q \Pi_l\| \le C \frac{\min((1+k)^m, (1+l)^m)}{(1+|k-l|)^2}.$$

For $a, b \ge 0$, $a + b = m$, the numerator is at most $(1+k)^a (1+l)^b$. Cauchy–Schwarz with the summable discrete kernel $(1+|k-l|)^{-2}$ therefore proves

$$|\langle Qu, v\rangle| \le C \|u\|_b \|v\|_a. \tag{26}$$

We use $(a, b) = (1/2, 1/2), (1/2, 3/2), (3/2, 3/2)$; no fractional chart interpolation is required.

For compact base support and zero fiber mean define

$$\mathcal{N}(\phi)^2 = \int_Y \{\|\nabla\phi\|_0^2 + \beta^2 \|\phi\|_1^2 + \beta^{-1} \|\nabla\phi\|_{1/2}^2 + \beta \|\phi\|_{3/2}^2\} \mathrm{d}V_\gamma. \tag{27}$$

Lemma 7.1. *Under (14), for large β_0 there is $c > 0$ independent of the shrinking parameter such that*

$$\|T\phi\|^2 - \|T^\flat \phi\|^2 \ge c \mathcal{N}(\phi)^2, \qquad T^\flat = \sum_i A_i(-\nabla_i + f_i B).$$

Proof. At degree $k \ge 1$, represent v by a harmonic homogeneous polynomial with Fischer norm $\|v\|_F^2 = \sum_\alpha \alpha! |v_\alpha|^2$. Its relation to probability sphere norm is $\|v\|_F^2 = N_k \|v\|^2$, $N_k = n(n+2)\cdots(n+2k-2)$. This follows by Gaussian integration by parts: trace-free coefficient tensors eliminate pairings within either factor, leaving precisely the Fischer sum. The radial moment satisfies

$\mathbb{E}|X|^{2k+2} = (n+2k)\mathbb{E}|X|^{2k}$. Put $d = k + n/2 - 1$ and $a_i = \partial_{z_i}$. For harmonic degree-k polynomials q_i,

$$\|\Pi_{k+1} \textstyle\sum_i z_i q_i\|_F^2 = \sum_i \|q_i\|_F^2 + \sum_{ij} (a_j q_i, a_i q_j)_F - d^{-1} \|\textstyle\sum_i a_i q_i\|_F^2, \tag{28}$$
$$\Pi_{k+1} \textstyle\sum_i z_i q_i = \sum_i z_i q_i - |z|^2 \sum_i a_i q_i/(2d).$$

Indeed $\Delta_z(|z|^2 h) = 4dh$ on degree $k-1$; multiplication and differentiation are Fischer adjoints. The probability-sphere adjoint is $A_i^* = (2d+2)\Pi_{k+1} z_i$ on degree k.

Write $D_i^{\pm} = \pm\nabla_i + k f_i$ and $C_{ij} = 2k(\mathrm{Hess}_\gamma f)_{ij}$. In the following calculation, $|\cdot|_F$ is the pointwise Fischer norm and $\|\cdot\|_F^2$ includes integration against $\mathrm{d}V_\gamma$. Put

$$s^{\pm}(y) = \left|\textstyle\sum_i a_i D_i^{\pm} v(y)\right|_F^2, \qquad S^{\pm} = \int_Y s^{\pm} \mathrm{d}V_\gamma,$$
$$X^+ = \int_Y \sum_{ij} (a_j D_i^+ v, a_i D_j^+ v)_F \mathrm{d}V_\gamma, \quad C[av] = \int_Y \sum_{ij} C_{ij} (a_i v, a_j v)_F \mathrm{d}V_\gamma.$$

Covariant integration by parts gives

$$X^+ = S^- - C[av] + O(k^2)\|v\|_F^2,$$
$$S^+ \le k\|D^- v\|_F^2 - C[av] + O(k^2)\|v\|_F^2,$$
$$\|D^- v\|_F^2 = \|D^+ v\|_F^2 + \int \mathrm{tr}\, C\, |v|_F^2.$$

Here $D_j^- D_i^+ - D_i^+ D_j^- = -C_{ij}$ plus curvature. A degree-k rotation has norm $O(k)$, and contractions add one factor k, giving the displayed error. Keep the input degree k fixed when commuting these base operators through polynomial derivatives. Substituting into (28) gives

$$N_k(\|Tv\|^2 - \|T^\flat v\|^2) = (2 + 2/d)\{d\|D^+ v\|_F^2 - S^+ - dC[av]\} + 2S^- + O(k^3)\|v\|_F^2. \tag{29}$$

Use the integrated bound on fraction $1-\theta$ of S^+ and on the remaining fraction use the pointwise bound, with $\zeta = c_1/\beta$,

$$s^+ \le (1-\zeta)(k+n-1)|D^+ v|_F^2 + \zeta\{2s^- + 8k^3\beta^2 |v|_F^2\}.$$

Integrate this pointwise bound with its displayed weights before combining with (29). The first contraction bound is at most $k+n-1$; the second follows from $D^+ + D^- = 2k\mathrm{d}f$ and $\sum_i \|a_i v\|_F^2 = k\|v\|_F^2$. The coefficient retained on $|D^+ v|^2$ is

$$(2 + 2/d)\{(n-2)/2 - \theta(n-1) + \theta\zeta(k+n-1)\}.$$

Choose $\theta > 0$ small in terms of n. This is bounded below by $c + c'k/\beta$. The S^- coefficient is nonnegative for large β_0. The remaining Hessian form is

$$-(2 + 2/d)\{(d - 1 + \theta)C[av] + (1-\theta)k \int \mathrm{tr}\, C|v|_F^2\},$$

which controls $k^3 \int \beta |v|_F^2$ for $n \ge 3$, $k \ge 1$. Take c_1 small to absorb its loss and then β_0 large to absorb curvature.

Both weighted cross terms have the needed sign. For $w \geq 0$,

$$\int w|D^+v|^2 = \int \{w|\nabla v|^2 + wk^2\beta^2|v|^2 - k\,\mathrm{div}(w\nabla f)|v|^2\}.$$

For $w = 1$, $\Delta f < 0$. For $w = \beta^{-1}$, $\mathrm{div}(\nabla f/\beta) = \beta^{-1}\,\mathrm{tr}_{(\nabla f)^\perp}\,\mathrm{Hess}\, f \leq -(n-1)c$. Thus the coefficient $c + c'k/\beta$ controls all four terms of (27). Divide by N_k and sum. Horizontal derivatives preserve degree and $T, T^\flat$ change it by one, so their distinct outputs are orthogonal. First sum finitely many degrees. At each fixed $\delta > 0$, smoothness and dominated convergence pass to the full norms; constants remain independent of δ, without a uniform harmonic tail cutoff. □

Lemma 7.2. *For sufficiently small ε and large β_0, $\mathcal{N}(\phi) \leq C\|P\phi\|$ for every compact-base-supported smooth ϕ with zero fiber mean.*

Proof. Take full adjoints in $\mathrm{d}V_\gamma \mathrm{d}\omega_y$. Direct degree adjunction and $[B, A_i] = -A_i$ give $Z = T^\flat - T^\dagger = \sum_i f_i A_i$. The exact squared-norm difference is

$$\|(2T+U)\phi\|^2 - \|(2T^\flat + U^\dagger)\phi\|^2 = 4(\|T\phi\|^2 - \|T^\flat\phi\|^2) + \langle \mathcal{E}\phi, \phi\rangle, \tag{30}$$
$$\mathcal{E} = 2[T^\dagger, U] + 2[U^\dagger, T] + [U^\dagger, U] - 2\{UZ + (UZ)^\dagger\}.$$

T has (base order, angular order, size) $(1,1,1)$ or $(0,2,\beta)$; U has $(a, 2-a, \varepsilon\beta^{-a})$, and Z has $(0,1,\beta)$. A scalar angular commutator lowers angular order by one; a base coefficient derivative lowers base order by one and costs at most β. The possible third-base commutators vanish because $U_2 = r\gamma^{ij}$ is angular independent. For $[U^\dagger, U]$, write $[U^\dagger - U, U]$; the real U_2 makes the first difference have base order at most one, and its possible third-base coefficient also commutes with U_2. Fourth-base terms are absent. The resulting types are

$$\varepsilon\beta^{-1}\Psi^1\partial_y^2, \qquad \varepsilon\Psi^2\partial_y, \qquad \varepsilon\beta\Psi^3.$$

One base integration by parts in the first type and (26) bound their forms by respectively $C\varepsilon\int \beta^{-1}\|\nabla\phi\|_{1/2}^2$, $C\varepsilon\int\|\nabla\phi\|_{1/2}\|\phi\|_{3/2}$, and $C\varepsilon\int\beta\|\phi\|_{3/2}^2$. For the middle type insert $\beta^{-1/2}$ and $\beta^{1/2}$ and use Cauchy–Schwarz. Coordinate/horizontal derivative differences are fixed angular order one and have the same bound. Thus $|\langle\mathcal{E}\phi,\phi\rangle| \leq C\varepsilon\mathcal{N}(\phi)^2$. Absorb this into Lemma 7.1 and discard the nonnegative comparison norm. Finally $\|J\phi\| \leq C\|\phi\|_1 \leq C\beta_0^{-1}\mathcal{N}(\phi)$ is absorbed after increasing β_0. □

Apply the last lemma to $\chi(\psi_\delta - 1)$. On the lower derivative support in (17), the density is the ordinary small-ball Poisson evaluation density. Lemma 6.1 makes it smooth down to $r = 0$, with uniformly bounded angular norms and first base derivative $O(1+r\beta)$. On $t > t_b$, radii have a positive lower bound and all profile derivatives are uniformly bounded; the fixed-separation kernel gives the same needed bounds. In particular, angular norms and $r\nabla\psi_\delta$ are bounded on these supports. Since $T1 = 0$, (25) gives a uniform bound for $P1$. The second-base term of $[P,\chi]$ is $r\gamma^{ij}(2\partial_i\chi\partial_j + \partial_i\partial_j\chi)$; the other terms have at most one base derivative, with coefficients $r\beta\Psi^1$, $r\Psi^0$, or bounded angular order one. The support bounds therefore give

$$\|P[\chi(\psi_\delta - 1)]\| = \|-\chi P1 + [P,\chi](\psi_\delta - 1)\| \leq C.$$

Coercivity proves the joint estimate

$$\sup_{0<\delta\leq 1}\|\chi\psi_\delta\|_{L^2(Y\times S_y, \mathrm{d}V_\gamma\mathrm{d}\omega_y)} < \infty. \tag{31}$$

This is not a pointwise-in-y density bound. On a fixed small Z with $\chi = 1$, $t < 0$, (20) gives for each matching pair

$$\|v_2 - v_1\|_{L^2(Z)} \le C_{v_1} \sup_Z r_\delta \|\psi_\delta\|_{L^2(Z \times S_y)} \longrightarrow 0.$$

The left side is independent of δ, so it is zero, and smoothness gives equality on Z. The same Z works for every pair; only the finite constant C_{v_1} changes. This completes Proposition 3.1.

8 Global equality from all cap sources

For real $F \in C_c^\infty(U_0)$ put $u_{j,F} = R_j F$. Define

$$\mathcal{Q} = \{p \in N^\circ : \text{ there is an open } V \ni p \text{ such that } u_{1,F} = u_{2,F} \text{ on } V \text{ for every } F\}.$$

The neighborhood precedes the universal source quantifier, so $\mathcal{Q}$ is open. Let W be its component containing the connected cap E, which belongs to $\mathcal{Q}$ by Lemma 2.2. Then $q_1 = q_2$ on W: outside $\overline{U_0}$ choose a source with nonzero value by Lemma 2.3 and subtract its two equations; on $\overline{U_0} \Subset E$ the coefficients already agree. For fixed $x \in W$, all-source equality makes $G_1(x,\cdot) - G_2(x,\cdot)$ vanish against sources on U_0. On W its delta singularities cancel because the local operators are now identical. Local regularity makes the difference a smooth homogeneous solution even at $y = x$. Unique continuation on connected W gives $G_1 = G_2$ on $W \times W$.

If W is proper in N°, it has an interior frontier. In a compact coordinate neighborhood, choose a point of W close enough to its complement that its Euclidean distance to that closed complement is smaller than its distance to the chart boundary. The inscribed ball is contained in W and touches its frontier at p. Its smooth boundary is a known one-sided hypersurface near p; no regularity of the whole frontier is required. Since $\overline{U_0} \Subset E$, p is outside it.

Choose one real F_0 with $u_{1,F_0}(p) = 1$. Equality from W and continuity give $u_{2,F_0}(p) = 1$. Set $s_j = u_{j,F_0}$ and choose one connected neighborhood Ω where they are positive, avoiding $\overline{U_0}$. For every F define $v_{j,F} = u_{j,F}/s_j$ there. Their equations are (4) and their pairs match on the known ball. Choose a finite set of n sources whose first-side ratio gradients span at p, using the arbitrary first jets; they remain spanning on a smaller Ω.

Proposition 3.1 gives a single connected neighborhood Z of p, meeting W, on which every ratio pair agrees. Subtract their normalized equations. The principal parts cancel, leaving

$$\langle \mathrm{d}\log s_1 - \mathrm{d}\log s_2, \mathrm{d}v_{1,F}\rangle_g = 0.$$

The spanning family gives $\mathrm{d}\log(s_1/s_2) = 0$ on Z. Matching on $Z \cap W$ makes its constant zero. Hence $s_1 = s_2$, $q_1 = \Delta_g s_1/s_1 = \Delta_g s_2/s_2 = q_2$, and all original source pairs agree on Z. Thus $W \cup Z$ is a connected subset of $\mathcal{Q}$, contradicting that p is on the frontier of its component.

The interior of a connected smooth manifold with boundary is connected: boundary half-balls meet only one interior component, so distinct interior components would give disjoint relatively open pieces partitioning N. Therefore $W = N^\circ$. The potentials agree on the original interior and on its boundary by smooth continuity. This proves Theorem 1.1.

AI use

This manuscript was prepared with assistance from OpenAI Codex in developing and examining candidate arguments, checking source statements, and drafting the exposition. The core continuation method is credited separately to its mathematical source. The manuscript is submitted for independent mathematical review.

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